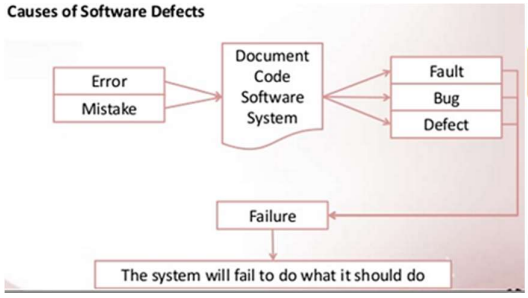
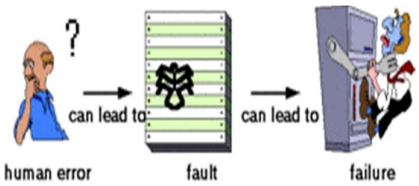
- Defect Life Cycle-
  - Difference between Bug, Defect, Failure, Error
    - Defect Tracking and Reporting
    - Types of Bugs, Identifying the Bugs, Reporting the Bugs
- **Introduction to automated testing-**
  - Install and configure selenium testing tool,
  - Case study: Design test case for login page, Internet Banking Login, Online shopping.

****

Difference between Bug, Defect, Failure, Error

"A coding mistake is called an **error**. The error found by the tester is called a **defect**. A defect accepted by the development team is called **bug**. If the software does not meet the requirements, then it is **failure**."





⦿ Error

An **Error** is a human mistake that produces an incorrect result. A mistake in the program leads to error due to incorrect loops or syntax. The mistake made by people. An Error appears not only due to the logical mistake in the code made by the developer but also anyone in the team can make mistakes during the different phases of software development.

➢ **Reasons of Error:**

- Misinterpret or misunderstand of software requirements.
- Miscalculation of some values in the coding.
- If a developer is unable to successfully compile or run a program.
- Mistakes in design or requirement activities.
- Customer may provide insufficient or incorrect information.
- Difference between actual and expected results.

➢ **Types of Error**

- User Interface Error: occurs during user interaction with the system. Such as missing or incorrect functionality of the system, no backup function etc.
- Syntax error: Misspelled words or grammatically incorrect sentences
- Calculation errors: occur due to bad logic, incorrect formulas, mismatched data type, etc.
- Flow control error: Occurs due to control of the program in wrong direction where software program acts unexpectedly i.e. infinite loop, syntax error during run-time, etc.
- Testing errors: Occur when implementing and executing the test process. i.e. bug scanning failure, inefficiency in reporting an error or defect.
- Hardware errors: Occurs due to configuration such as insufficient storage area for memory and no compatibility with the device.

⦿ Fault

Fault is a condition that causes a system to fail in performing its required function. Fault is incorrect step, process or data definition in a computer program which causes program acts an unexpected manner which is not in the desired output. It is an evidence of the error.

⦿ Bug

"A bug is an error, weakness, mistake or fault present in a coding that produces an incorrect or unexpected result that impact on the performance as well as functionality of the software". Bugs arise when mistakes and errors made by people in either source code or its design. Typically bugs are found in unit testing by testers.

- **Types of Bugs**
  - Algorithmic bugs such as division by zero.
  - Logic bugs such as Infinite loops (a loop having no termination point).
  - Resource bugs due to the use of uninitialized variables.
  - Functional Errors
  - Compilation and Runtime Errors
  - Missing commands and Error handling errors

⊙ Failure

Once the product is delivered to the customers and if the customer find any issues in the software it is called as condition of failure. Failure occurs when executing software code, that time software code fails to meet software requirements or functions due to occurrence of any defect i.e. Failure occurs when fault executes.

- **Reasons for Failures:**
  - Human Error: Human interacting with the software by providing wrong or incomplete inputs.
  - Environmental Conditions: Failures also happen by environmental conditions like a radiation burst, overheating, electronic fields, or pollution could cause faults in hardware or software.
  - Users: Failures may also occur in the system if the user tries to execute operations with the intention of breaking the system.

Example1:

You are driving a car and you are on road while on driving now there is two way on the road
1) Left--> Mumbai
2) Right--> Delhi

You have to go to Delhi. you have to turn the steering to the right, but by mistake you turn the steering to the left, from that position that is called as "Error" because human interaction is there and now Fault is there till you will not reach the Mumbai, but when you reach Mumbai that is a final stage which is called "Failure" because you had to reach Delhi but now you are in Mumbai.

Example 2: this program is required to add two numbers.

#include<stdio.h>
int main ()
{
int value1, value2, ans;
value1 = 5;
value2 = 3;
ans = value1 - value2;
printf("The addition of 5 + 3 = %d.", ans);
return 0;
}
	#include<stdio.h>
	int main ()
	{
	int value1, value2, ans;
	value1 = 5;
	value2 = 3;
	ans = value1 - value2; // ----> Bug
	printf("The addition of 5 + 3 = %d.", ans);
	return 0;
	}

After compile and run program o/p is: **The addition of 5 + 3 = 2.**

So after output of program, we realize the program has failed. Because program displays the answer 2 instead of 8. We have detected a bug. Analyze the program what was the fault in the program.

We notice that a '-' sign present instead of '+' sign. So the fault in the program is the '-' sign. A mistake made by typing symbol '-' instead of '+' sign. Failure occur in execution of the program. We found bug. When tester testing the functionality of the program that time he realize output is faulty and raise a defect. Due to wrong result it is known error which resulted in the fault in the program and due to which the program failed to give the correct result.

⊙ Defect

The variation between the actual results and expected results is known as defect. A defect is an error in coding or logic that causes a program to fail or produce incorrect result. A bugs introduced by programmer inside the code and corrected by the developer in development phase.

When a tester finds a defect they need to documented the details of the defect and communicate to the development team. A report contains the list of defects that have been found by

the tester, along with the details of the defects. This communication can take place through email by reporting the defect through a defect tracking tool like JIRA, Bugzilla etc.

➢ **Reasons of Defects:**

- User providing wrong and inaccurate inputs.
- Analyst interprets requirements incorrectly.
- Incorrect design.
- Error in coding.
- Data entry error.

➢ **Types of Defects:**

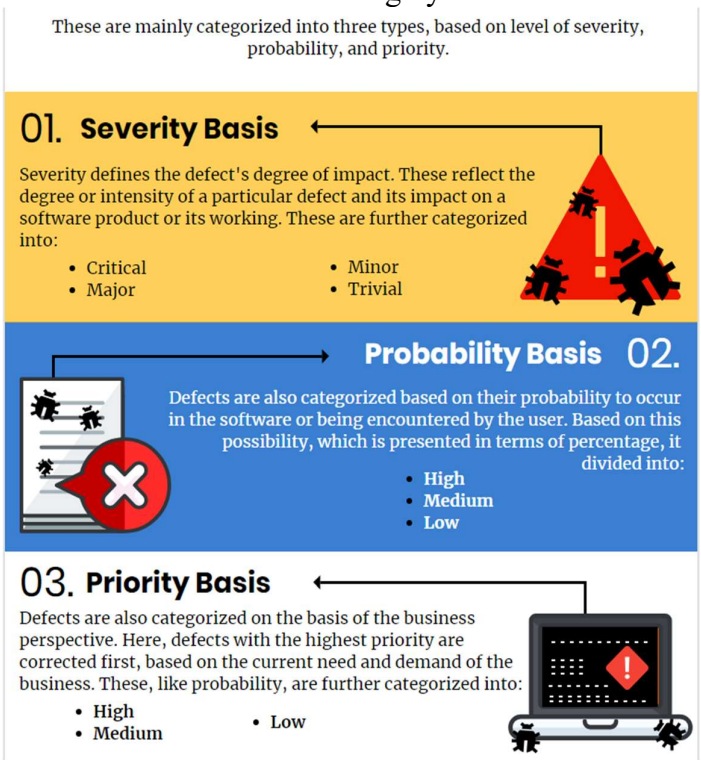
- **Missing Defects:** Occur due to client requirements not satisfy.
  Example: if we take the above example and open the inbox link, we see that it is not there only, which means the feature is not developed only.
- **Wrong Defects:** Occur due to incorrect implementation of requirements. Which not mentioned in the requirement specification.
  Example: Suppose a Gmail application, where we click on the **"Inbox"** link, and it navigates to the **"Draft"** page, this is happening because of the wrong coding which is done by the developer, that's why it is a wrong defect.
- **Extra Defect:** It is caused by a feature not in original list of client requirements but added later into the software.
  Example: Suppose we have one application form wherein the **Name field**, the **First name**, and **Last name** textbox are needed to develop according to the client's requirement.
  But, the developers also develop the **"Middle name"** textbox, which is not needed according to the client's requirements

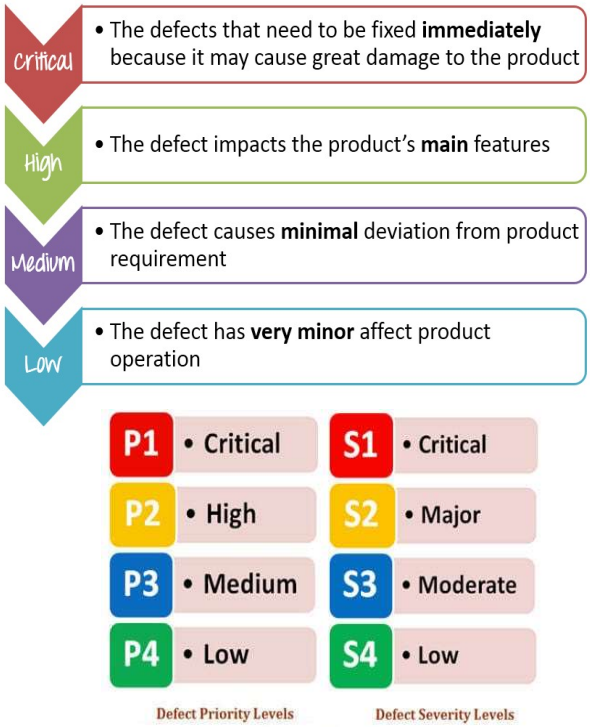
➢ **Benefits of using a defect report**

- Defect report helps the development team (and testing team) to understand the defect
- It also enables the teams to track the defect to closure
- If the defect is not reported, which may lead to the defect not being fixed

➢ **Defect categorization**

Defects are classified from the QA team view as **Priority** and from the development view as **Severity** (complexity of code to fix it). These are two major classifications that play an important role in the timeframe and the amount of work that goes in to fix defects on priority basis. i.e. fixing those defects first that are highly critical.





No	Description	Priority	Explanation
1	The website performance is too slow	High	The performance bug can cause huge inconvenience to user.
2	The login function of the website does not work properly	Critical	Login is one of the main function of the banking website if this feature does not work, it is serious bugs
3	The GUI of the website does not display correctly on mobile devices	Medium	The defect affects the user who use Smartphone to view the website.
4	The website could not remember the user login session	High	This is a serious issue since the user will be able to login but not be able to perform any further transactions

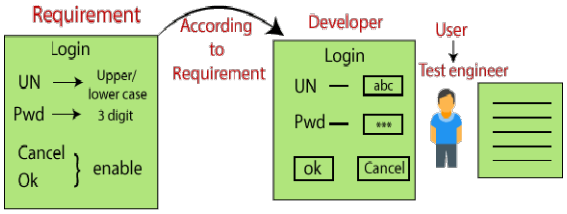
Defect/Bug Report or Defect Tracking

Defect Report is a document where Test lead reported bug to developer. Using this documents developer identify exact position as well as location of the defect, so he can fix it easily.

- **Defect Id** – The Unique identification number assigned to the defect / bug
- **Defect Description** – Detailed description of the Defect
- **Date Raised/Reported on** – Date when the defect is reported.
- **Detected By/Reported by** – Name of the person who reported defect.
- **Assigned To/Fixed by** - Name/ID of the developer is assigned to analyze/ fix the defect.
- **Severity** – Impact of the defect or bug. This can be minor, major or critical.
- **Priority** – This can be low, medium or high. It is the priority or importance of the defect.
- **Status** –Track the status of the defect. The status can be New, Assigned, Open, Rejected, Fixed, Differed, Reopened, Verified or Closed.
- **Version** – Version of the application in which defect was found.
- **Closed On** – Date when the defect is closed

Defect ID	Boo12
Module	Login
Test case name	Gmail_login_Compose mail
Reported by	Kundan Test engineer
Version	3.0
Status	assigned
Date	23-02-2020
Assign to	Malvika developer
Severity	critical
Priority	P2
Platform	Window XP, internet explorer7.0
Build no.	B02
Test data	Username=xyz, password= 123
Brief description	
Navigation steps	▪ Login Gmail application ▪ Compose mail and confirmation message should be displayed
Expected result	Should also be in the inbox.
Actual result	Mail not found in inbox

Example:

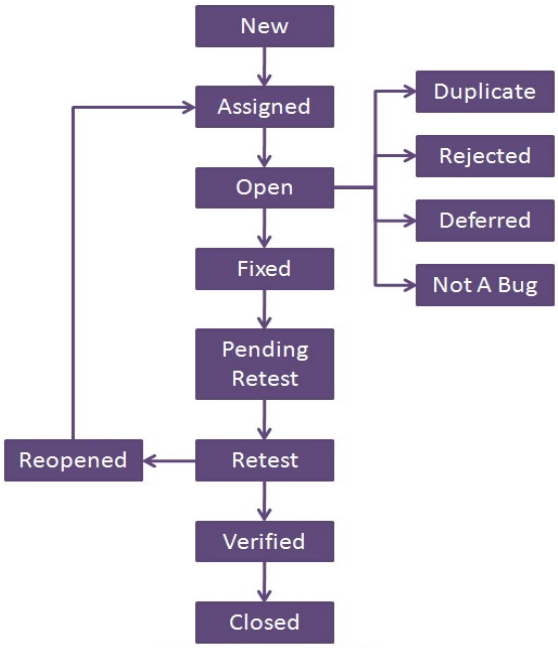


Defect/Bug Life Cycle

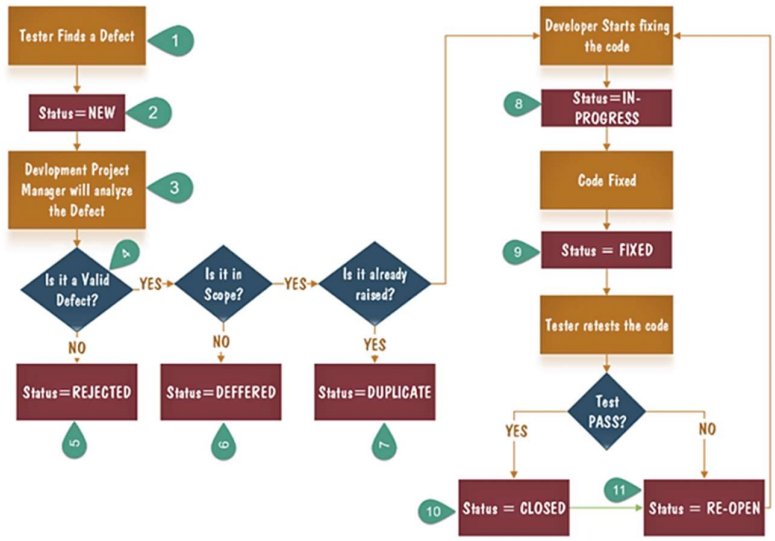
No software exists without a defect or bug. Removing bugs from the software depends upon the efficiency of testing on the software. A defect is the difference between the expected result and the actual result. The cost of finding and correcting of defects is most expensive software development activities.

Defect life cycle is a process of report and resolve also track the status of bugs. The goal of defect life cycle or bug life cycle is to resolve and test the bug or defect and follow the process till defect or bug exists. Defect or bug can be raised at any point during testing and could be reported to the relevant development team to fix it. Defect fixing process systematic and efficient.

Defect life cycle is starts when defect is found and ends when a defect is closed, after ensuring it's not reproduced. Defect goes through number of states.



Bug / Defect Lifecycle



New:	When a bug is detected for the first time in the software. It is assigned a status NEW.
Assigned:	A Reported bug testing team will approved the defect is genuine and assigned to the corresponding developer for fixing.
Open:	The developer starts analyzing and works on the defect fix
Fixed:	When a bug is resolved by the developer after making necessary changes in code
Pending retest:	After the developer has fixed up the defect, they send an application to the tester for testing. Retesting remains pending from the testers end
Retest:	After the defect is fixed by developer, it is assigned to the tester for retest the application again to see if the bug is reproduced again.
Verified:	If tester does not find any type of bug in the software, then he approves that the bug is fixed and changes the status to "Verified".
Reopen:	If the bug still exists even after the bug is fixed by the developer, the tester changes the status to "Reopened". The bug traverses the life cycle once again.
Closed:	Once the bug is fixed, it is tested by the tester. If the tester feels that the bug no longer exists in the software, he changes the status of the bug to "Closed".
Duplicate:	If the bug was reported earlier and again it is reported then the status is "Duplicate".
Rejected:	If the developer feels that the bug is not genuine, he rejects the bug. Then status is changed to "Rejected".
Deferred:	If the reported bug is not a higher priority and it can be fixed in the next release then the defect is marked as "Deferred" and will be fixed in upcoming releases.
Not a bug:	If testing team finds out that the defect is not harmful to the application then the status assigned to "Not a bug".

Defect/Bug Report Example: Creating User Module of banking Application	
Bug ID:	B001
Bug Name:	Application crash on clicking the SAVE button while creating a new user.
Description:	Application crash on clicking the SAVE button while creating a new the user, hence unable to create a new user in the application.
Build:	Version Number 5.0.1
Severity:	S1 Critical
Priority:	P2 High
Reported By:	Nayan-Tester
Assigned to:	Ishan-Developer
Reported On:	02 January 2021
Status:	New
Environment:	Windows 10, MYSQL
Area Path:	USERS menu > New Users
Steps To Reproduce:	**1)** Login into the Application **2)** Navigate to the Users Menu > New User **3)** Filled all the user information fields **4)** Clicked on the 'Save' button **5)** Seen an error page "ORA1090 Exception: Insert values Error…" **6)** See the attached logs for more information **7)** And also see the attached screenshot of the error page.
Expected Result:	On clicking SAVE button, should be prompted to a success message "New User has been created successfully".
Actual Result:	On clicking SAVE button, show 'application crash'

Introduction to automated testing-

Automation Testing or Test Automation is a software testing technique that performs using special automated testing software tools to execute a test case suite. This can be achieved by writing test scripts or using any automation testing tool. Test automation is used to automate repetitive tasks and other testing tasks which are difficult to perform manually. **Test Automation** is the best way to increase the effectiveness, test coverage, and execution speed in software testing.

- **The test automation performs testing at three different levels:**
  1. **Unit Level Automation**
  2. **API Testing**
  3. **User Interface**
- **Types of Automation with Tools:**
  - **Functional Test Automation**
    Automating Software Application Functionality Test Scenarios using any supported Software Test Tool, like UFT/QTP, Selenium, RFT, SilkTest, TestNG etc. Selenium is open source tool.
  - **Performance Test Automation:**
    Automating Software Application Load, Stress, Spike, Data Volume etc.. Test Scenarios using Software Test Tool like LoadRunner, RPT, Silk Performer, JMeter etc
  - **Test Management**
    Test Management is an activity of managing the computer software testing process. (Requirements management, Test Management (Manual & Automated), and Defect management)
    ALM / QC, Jira, QAComplete…)

- **Advantages of Automation Testing**
  1. Increases amount of test coverage
  2. Speeds up the testing process
  3. Quality Assurance and Safety Assurance
  4. Improves Reliability of Tests

5. Reduces the test execution time and cost.
6. Minimizing Human Interaction
7. Automation scripts can run at any time and any number of times.
8. Reusability of scripts on other applications.
9. Easy to test Stress, Performance and Load
10. Saves manual efforts.

➢ **Disadvantages of Automation Testing**

1. Debugging the test script is a major issue. It may lead to consequences if any error is present in the test script
2. Proficiency is required to write automation test scripts
3. Test maintenance is costly
4. Maintenance of test data is difficult.
5. Testers should have basic programming knowledge
6. Testers should have experience in writing automation test scripts.

➢ **Which Test Cases can be automated?**

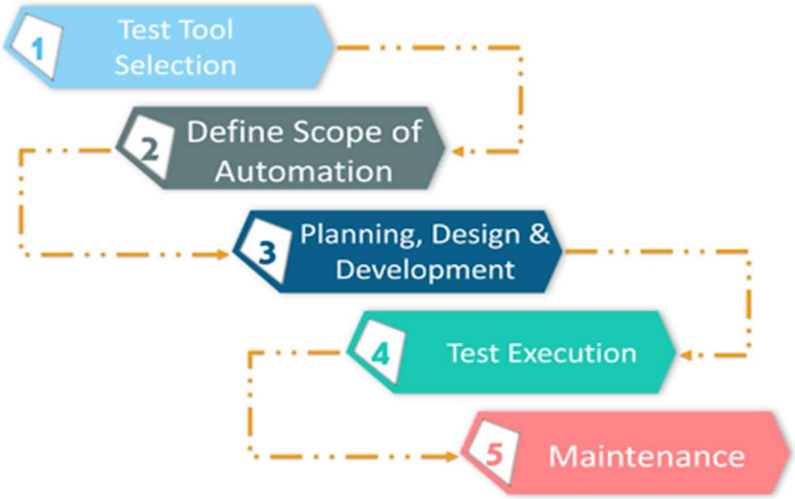
Test Cases should be automated if the following conditions are met.

1. Tests that are used repeatedly
2. High-Risk test cases
3. Tests that tend to cause human error
4. Test Cases that are impossible to perform manually
5. Tests that take lot of time to do manual testing
6. Time consuming test scenarios
7. Regression test suites
8. GUI items
9. Database connections.

➢ **Automated Testing Process:**

Following steps are followed in an Automation Process

Step 1) Test Tool Selection
Step 2) Define scope of Automation
Step 3) Planning, Design and Development
Step 4) Test Execution
Step 5) Maintenance



Install and configure selenium testing tool

The key-step to begin with Selenium is to install and get familiar with it. In order to help you out, I have come up with an article on Selenium Installation. Through the medium of this article, I will guide you through the simple steps to configure and install selenium on your system.

The process of installing Selenium involves 3 steps, namely:

1. Install Java
2. Install Eclipse IDE
3. Install Selenium WebDriver

Let's get started.

Selenium supports various programming languages like Java, Python, C#, Perl, Ruby etc. Java is one of the most widely used languages for Selenium. Let's begin and understand the process of Java Installation. The very first step, needed for Java installation is to configure Java environment so that later on Java codes can be used to drive the selenium.

Install Java

Follow below steps to complete your Java installation.

- Go to the Java Downloads Page and click on the option for *Java Platform (JDK)*.

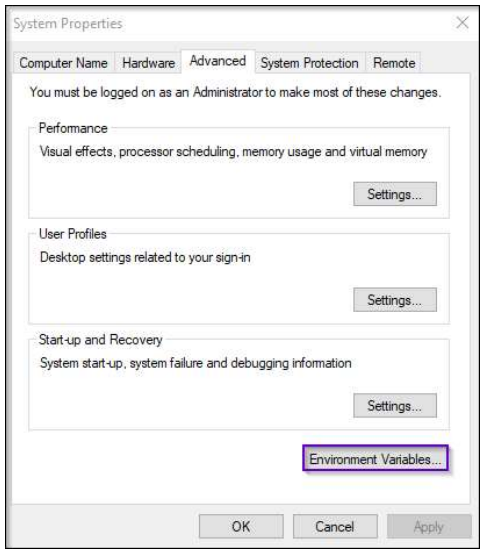
Java SE Development Kit 11.0.1

You must accept the Oracle Technology Network License Agreement for Oracle Java SE to download this software.

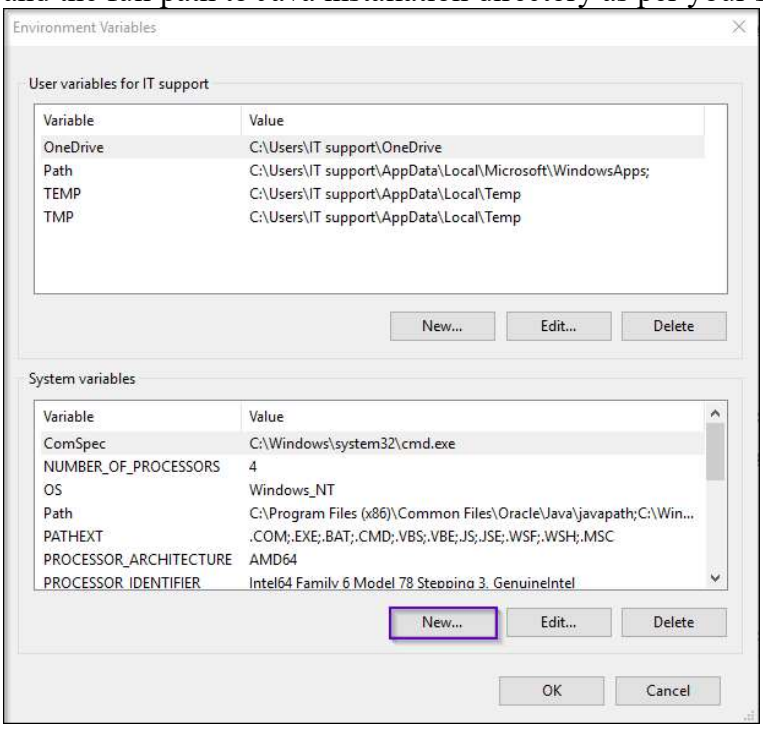
Accept License Agreement Decline License Agreement

Product / File Description	File Size	Download
Linux	147.4 MB	jdk-11.0.1_linux-x64_bin.deb
Linux	154.09 MB	jdk-11.0.1_linux-x64_bin.rpm
Linux	171.43 MB	jdk-11.0.1_linux-x64_bin.tar.gz
macOS	166.2 MB	jdk-11.0.1_osx-x64_bin.dmg
macOS	166.55 MB	jdk-11.0.1_osx-x64_bin.tar.gz
Solaris SPARC	186.8 MB	jdk-11.0.1_solaris-sparcv9_bin.tar.gz
Windows	150.98 MB	jdk-11.0.1_windows-x64_bin.exe
Windows	170.99 MB	jdk-11.0.1_windows-x64_bin.zip

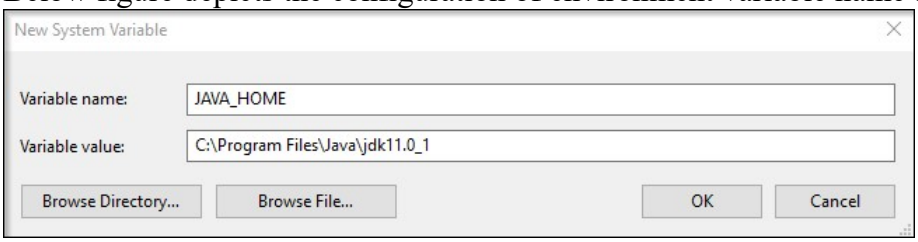
- In the next page, select the Accept License Agreement radio button, accept it and click the download link against your matching system configuration.
- You can run the installer once the download is over and follow onscreen instructions.
    - Go to start and search for 'System'
    - Click on 'System'
    - Click on 'Advanced system settings'
    - Click on 'Environment Variables' under 'Advanced' tab as shown below:



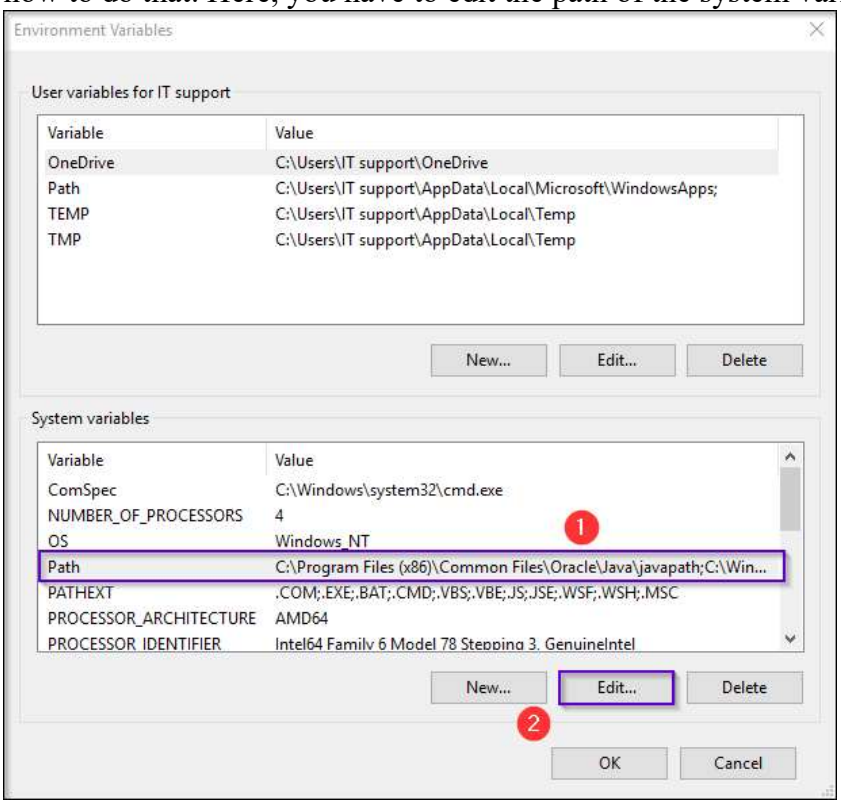
- Next, under system variables choose new and enter the variable name as 'JAVA_HOME' and the full path to Java installation directory as per your system as shown below:



Below figure depicts the configuration of environment variable name and value.



- Next thing that you have to do is to configure your environment variables. Let's see how to do that. Here, you have to edit the path of the system variable as shown below.



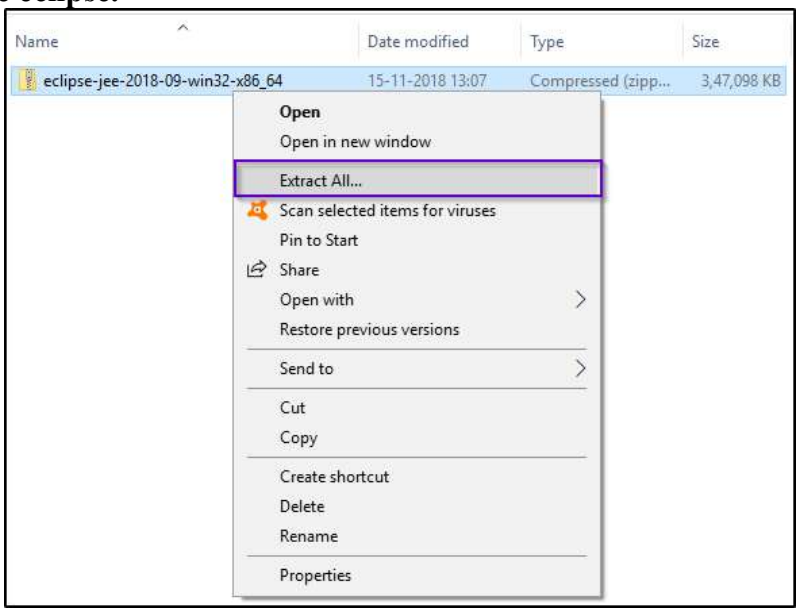
- Under 'Variable value', at the end of the line, enter the following path – *%JAVA_HOME%bin;*
  Now, you can click 'OK' and you are done.
- Now to cross-check the installation, just run following command in cmd – *java -version*. It should display the installed version of Java in your system.

Install Eclipse

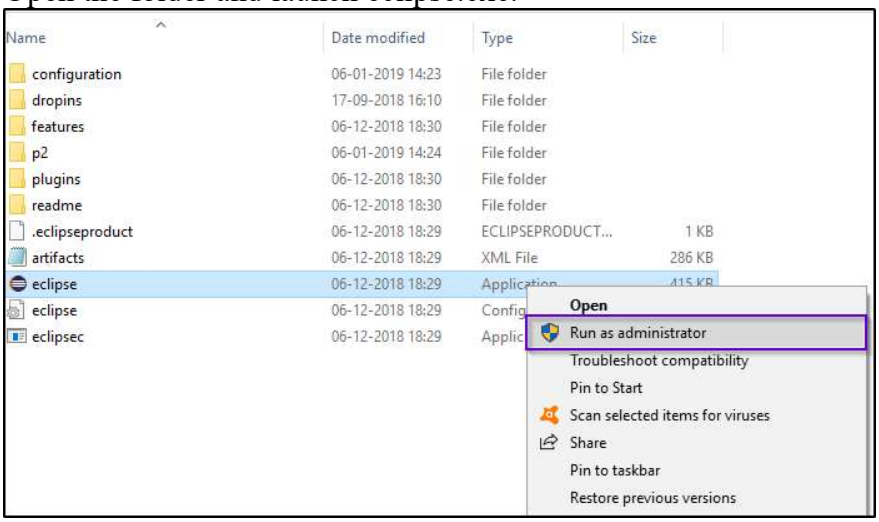
Follow the below steps to configure Eclipse on your system:

- Navigate to the following URL – https://www.eclipse.org/downloads/ and select the download link depending on your system architecture – (32 Bit or 64 Bit) and download it.

- **Once the download is over, extract the zipped file and save it to any directory. The root folder is the eclipse.**



- Open the folder and launch eclipse.exe.



Now, the last step is to install Selenium.

Install selenium and configure it in your system.

Install Selenium Web Driver

- Open the browser and navigate to http://www.seleniumhq.org.
- Click 'Download' menu and choose 'Download version'. i.e. x.y.z

  - The download should get started for 'selenium-server-standalone-x.y.z.jar'. Save this JAR in "C:Selenium".
  - Next, you have to download Selenium Java Client. In the same downloads page, scroll down and you will find a section called Selenium Client and WebDriver **Language.**

Now, click on Download link under Java as shown below.

Selenium Client & WebDriver Language Bindings

In order to create scripts that interact with the Selenium Server (Selenium RC, Selenium Remote WebDriver) or create local Selenium WebDriver scripts, you need to make use of language-specific client drivers. These languages include both 1.x and 2.x style clients.

While language bindings for other languages exist, these are the core ones that are supported by the main project hosted on GitHub.

Language	Client Version	Release Date			
Java	3.141.59	2018-11-14	Download	Change log	Javadoc
C#	3.14.0	2018-08-02	Download	Change log	API docs
Ruby	3.14.0	2018-08-03	Download	Change log	API docs
Python	3.14.0	2018-08-02	Download	Change log	API docs
Javascript (Node)	4.0.0-alpha.1	2018-01-13	Download	Change log	API docs

- Extract the zip file and save it in your selenium folder.
- Next, you have to download Chrome Driver for Google Chrome and Gecko Driver if you are using Mozilla Firefox.
- In the same downloads page, scroll further down and you will find third-party drivers section. There you will find options for downloading gecko driver and chrome driver as shown in the below figure.

Third Party Drivers, Bindings, and Plugins

Selenium can be extended through the use of plugins. Here are a number of plugins created and maintained by third parties. For more information on how to create your own plugin or have it listed, consult the docs.

Please note that these plugins are not supported, maintained, hosted, or endorsed by the Selenium project. In addition, be advised that the plugins listed below are not necessarily licensed under the Apache License v.2.0. Some of the plugins are available under another free and open source software license; others are only available under a proprietary license. Any questions about plugins and their license of distribution need to be raised with their respective developer(s).

Third Party Browser Drivers NOT DEVELOPED by seleniumhq

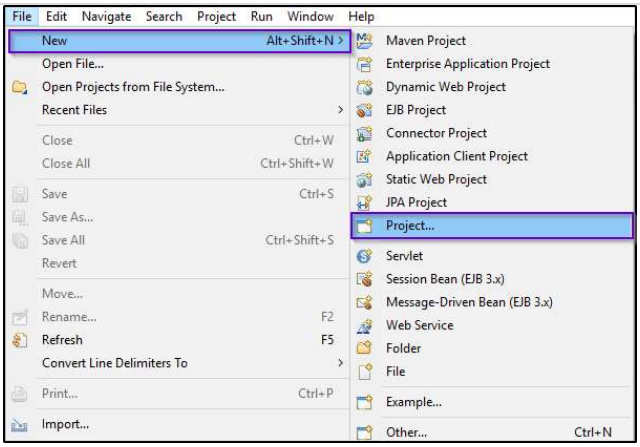
Browser				
Mozilla GeckoDriver	latest	change log	issue tracker	Implementation Status
Google Chrome Driver	latest	change log	issue tracker	selenium wiki page
Opera	latest		issue tracker	selenium wiki page
Microsoft Edge Driver			issue tracker	Implementation Status

- In the next page, you will find the latest version available. You can choose that and download based on your system configuration. You can refer the below image for your OS configuration.

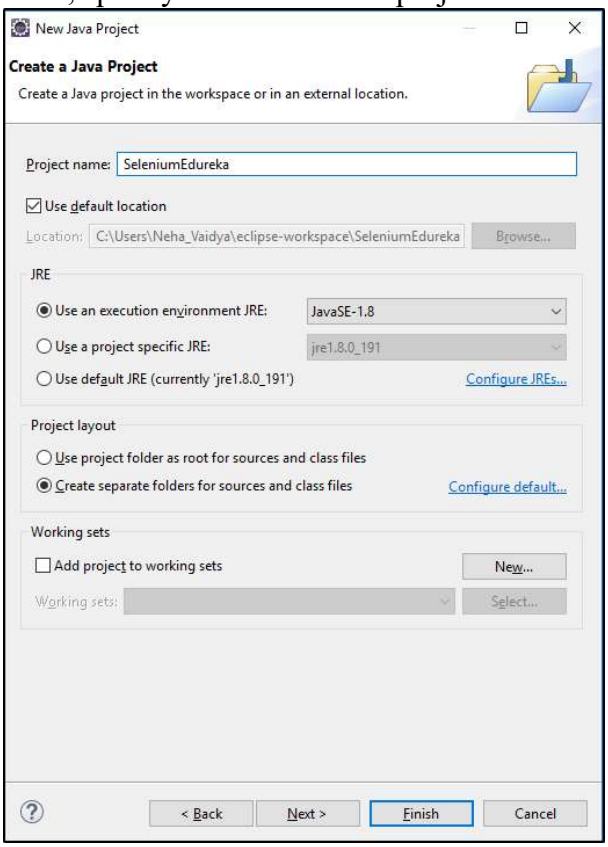
Index of /2.45/

Name	Last modified	Size	ETag
Parent Directory		-	
chromedriver_linux64.zip	2018-12-10 23:20:22	5.14MB	61c3484abc908c543b823cf42f58fc5d
chromedriver_mac64.zip	2018-12-11 04:12:22	6.63MB	9f6b235f30b6ea4579612273595c8301
chromedriver_win32.zip	2018-12-11 01:10:40	4.39MB	6ee964d7172fb8bcd8538ccb430b53ed
notes.txt	2018-12-11 05:17:39	0.02MB	d98475f7fad0a16aa0543c01defa518c

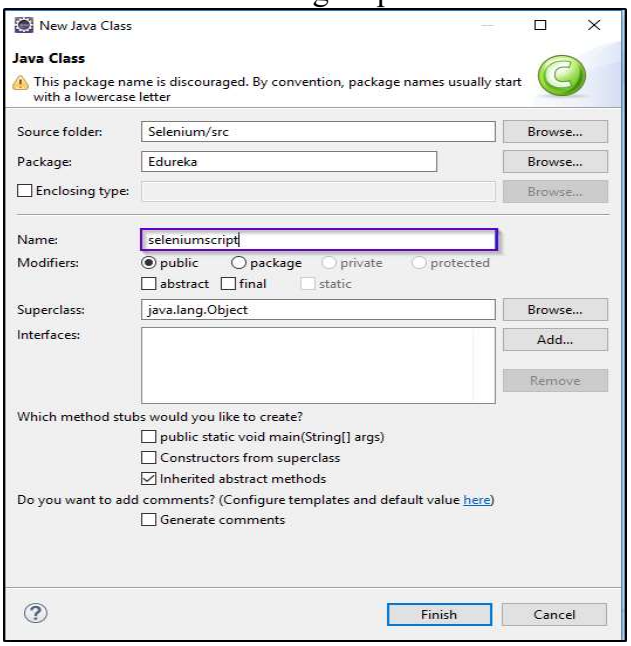
- You can simply download the driver, extract the zip file and save it in any location. Similarly, you can download the latest version of gecko driver and save it if you are using the Mozilla browser.
- Next, you can simply create a new project in Eclipse IDE and add all the external selenium jar files from selenium lib directory.
    - First, launch Eclipse and create new project as shown below.



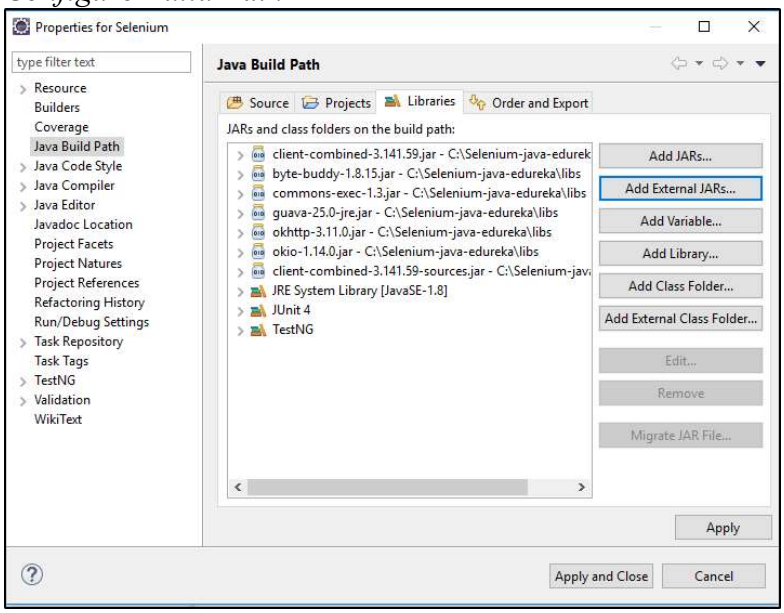
Next, specify the name of the project and create a class.



Let's create a class file and give public access modifier.



- Next, you should add referenced libraries to your project. Click on your *project -> Build Path -> Configure Build Path* and add external JARs as shown in the figure.



- **Now you are all set to execute the first Selenium program. You can run the below script and check the output.**

```
package edureka;
import java.util.concurrent.TimeUnit;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
 public class FirstSeleniumScript {
public static void main(String[] args) throws InterruptedException{
System.setProperty("webdriver.chrome.driver", "C:Selenium-java-
edurekachromedriver_win32chromedriver.exe");
WebDriver driver = new ChromeDriver();
driver.manage().window().maximize();
driver.manage().deleteAllCookies();
driver.manage().timeouts().pageLoadTimeout(40, TimeUnit.SECONDS);
driver.manage().timeouts().implicitlyWait(30, TimeUnit.SECONDS);
driver.get("https://login.yahoo.com/");
driver.findElement(By.xpath("//input[@id='login-
username']")).sendKeys("edureka@yahoo.com");
}
}
```

Write a Test case for Facebook Login page Functionality using TestNG.

```
package FacebookLogin;
import org.openqa.selenium.By;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.testng.Assert;
import org.testng.annotations.Test;

public class FacebookTest
```

```
{
    @Test
    public void verifyFacebookLoginPage() throws InterruptedException
    {
            System.out.println("Hello Login into Facebook");
    System.setProperty("webdriver.chrome.driver","C:\\browserdriver\\chromedriver.exe");
            ChromeDriver driver=new ChromeDriver();
           driver.get("https://www.facebook.com");
   driver.manage().window().maximize();
            WebElement LoBtn=driver.findElement(By.xpath("//button[text()='Log In']"));
            String LoginButton=LoBtn.getText();
            Assert.assertEquals(LoginButton, "LogIn");
            Thread.sleep(9000);
            System.out.println("Close Browser");
            driver.close();
    }
    @Test
    public void verifyFBLoginWithValidCredentials() throws InterruptedException
    {
            System.out.println("Hello Login into Facebook");
    System.setProperty("webdriver.chrome.driver","C:\\browserdriver\\chromedriver.exe");
            ChromeDriver driver=new ChromeDriver();
            driver.get("https://www.facebook.com");
            driver.manage().window().maximize();
driver.findElement(By.xpath("//input[@class='inputtext_55r1_6luy']")).sendKeys("sangolacollege
@gmail.com");
            driver.findElement(By.xpath("//*[@id=\"pass\"]")).sendKeys("sangola");
            driver.findElement(By.xpath("//button[text()='Log In']")).click();
            Thread.sleep(17000);
            System.out.println("Close Browser");
            driver.close();
            }
}
```